OB+RCT
CATE/Overlap/High dimension/Matching

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Overview*

- why combine rct and ob (poor overlap in ob)
- review of overlap assumption in causal inference (IPW, PLM, R-learner, CRF)
- measure of overlapness and balance (served as a first step)
- review of overlap/balance intuited method
- general question: how can rct be better combined with ob (in a 'better overlapness/balance therefore better statistical properties' sense)?[1]

[1] A side question can be: if we have ob data, can we better design a random experiment?

why combine rct and ob, and super-population framework

combine rct and ob - why?

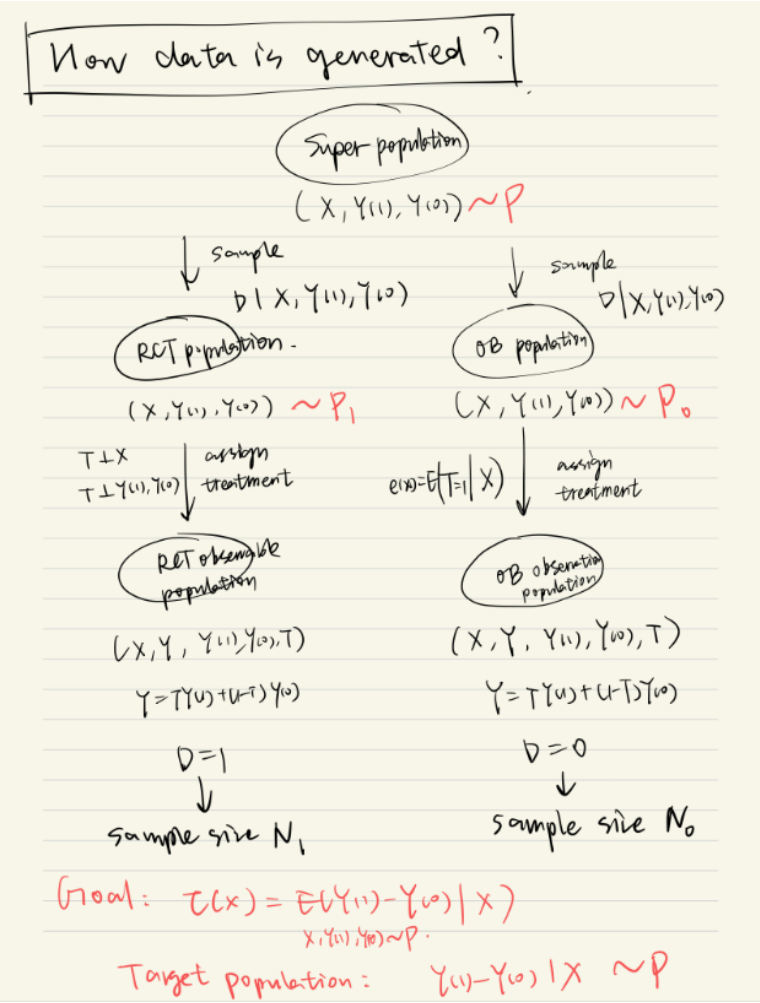
- RCT+EC
  - RCT so small
  - RCT may only cover some specific patient(X support region)
- OB+RCT
  - OB with poor overlap(in what sense?) our case
  - OB with unmeasured confounding

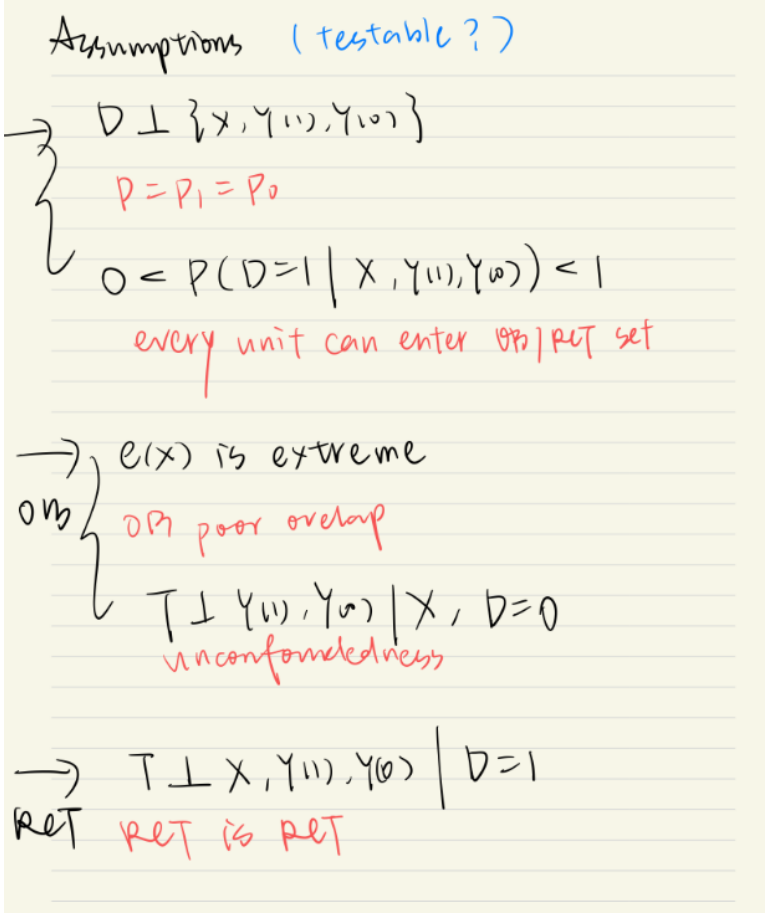
Poor overlap in OB*

- why poor overlap?
  - highly but not totally deterministic assignment mechanism(e.g. medical interventions often based on guidelines)
  - side question: is the assignment mechanism good (propensity is high when effect is high)? (e.g. is the guideline optimal for patient?) Would that affect how we improve the current guideline?
- when does poor overlap matter?
  - the unbalance of confounders has more influence
  - the complexity of propensity/outcome model?
- poor overlap, then?
  - need more data, conduct rct(how much rct samples do we need? better ways to get rct?)
  - how to use these new rct data?

combine rct and ob - a data generating view

In super population framework, both RCT and OB are samples from a stochastic distribution. (not finite population and design-based experiment framework) (Liu, L. Thomas, et al. 2025) introduce a double-indexed notation.





review of the role of overlap assumption

OB - why overlap important

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)
- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for(part of properties) statistical analysis of many estimators

Overlap in IPW

- oracle IPW estimator[2] is asymptotically normal

$$\sqrt{n}(\hat{\tau}^*_{ipw} - \tau) \sim N(0, V^*_{IPW})$$

where the asymptotical variance

$$V_{IPW^*} = \mathrm{Var}\left[\tau(X_i)\right] + \mathbb{E}\left[\frac{\left(\mu_{(0)}(X_i) + (1 - e(X_i))\tau(X_i)\right)^2}{e(X_i)(1 - e(X_i))}\right] + \mathbb{E}\left[\frac{\sigma^2_{(1)}(X_i)}{e(X_i)} + \frac{\sigma^2_{(0)}(X_i)}{1 - e(X_i)}\right]. \quad (1)$$

- all moments need to be well-defined and finite
  - if potential outcomes are uniformly bounded, it means we need $E(\frac{1}{e(X)(1-e(X)}) < \infty$
  - if the potential outcomes have finite second moments, it means we need bounded propensity, i.e. strict overlap

[2] unbiasedness with balancing property $E(XZ/e(X)) = E(X(1-Z)/(1-e(X))$

Overlap in PLM*

Consider Partial linear model

$$E(Y \mid X, Z) = \beta^T Z + \theta(X)$$

- (Robinson 1988) (generalized FWL theorem) regression-of-residuals estimator is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if
  - $\Phi = E[(Z - E(Z|X))(Z - E(Z|X))^\top]$ is positive definite.
    - $\Phi^{-1} = \frac{1}{E(e(x)(1-e(x))} = \frac{1}{E(Var(Z|X))}$ (convergence rate & overlap)

Overlap in R-learner

Considering HTE, now the PLM model becomes:

$$E(Y \mid X, Z) = \beta(X)Z + \theta(X)$$

- (Nie and Wager 2021) equivalently

$$\tau^*(\cdot) = \arg\min_{\tau} \mathbb{E}\Big[\big(Y - m^*(X) - (W - e^*(X))\tau(X)\big)^2\Big]$$

  therefore estimator(minimize R-loss, cross fitting):

$$\hat{\tau} = \arg\min_{\tau} \big(\hat{L}_n(\tau) + \Lambda_n(\tau)\big)$$

  - penalized kernel R-learner is near oracle(depends on the local robustness property of R-loss) if
    - strict overlap: turning regret bound into MSE bound

$$(1-\eta)^{-2}R(\tau) < E[\{\tau(X_i) - \tau^*(X_i)\}^2] < \eta^{-2}R(\tau)$$

    - RKHS, good estimate of nuisance functions, kernel, ...
  - when $e(X) \approx 0$, R-learner roughly equivalent to X-learner(?)

Overlap in CRF - high-dimensional/heterogeneity*

(Wager and Athey 2018)

- Bias of estimation

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular

more on overlapness and balance

OB - overlap & balance

Define P_1 by $\frac{dP_1}{dP} = \frac{e(X)}{E_p(e(X))}$ and P_0 by $\frac{dP_0}{dP} = \frac{1-e(X)}{E_p(1-e(X))}$, which are the distribution of covariates in treatment and control arm. Let $P(Z=1) = \pi$ be a constant $\pi \in (0,1)$.
By Bayes' theorem, we have density ratio(a measure of balance) with propensity ratio(a measure of overlap) as

$$\frac{dP_1}{dP_0} = \frac{P(X|Z=1)}{P(X|Z=0)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{P(Z=0)}{P(Z=1)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{1-\pi}{\pi}$$

and the mean discrepancy(a measure of balance) is

$$E(X|Z=1) - E(X|Z=0) = E_{P_1}(X) - E_{P_0}(X)$$
$$= E_P(X(\frac{dP_1}{dP} - \frac{dP_0}{dP})) = E_P(X(\frac{e(X)-\pi}{\pi(1-\pi)}))$$

Also can derive other f-divergence form.

$$D_f(Q_1 \| Q_0) := E_{Q_0}[f(\frac{dQ_1}{dQ_0})]$$

OB - strict overlap & balance in high-dimensional setting*

- Strict overlap with η implies bounded density ratio.

$$b_{min} =: \frac{1-\pi}{\pi}\frac{\eta}{1-\eta} \leq \frac{dP_1}{dP_0} \leq \frac{1-\pi}{\pi}\frac{1-\eta}{\eta} := b_{max}$$

- Strict overlap restricts global discrepancies between P_0 and P_1([support same?]), and these restrictions become more restrictive as the dimension of covariates grows large(D'Amour et al. 2021)[3]

$$D_f(P_1 \| P_0) \leq B_f(b_{min}, b_{max})$$

$$D_f(P_0 \| P_1) \leq B_f(b_{max}^{-1}, b_{min}^{-1})$$

- From data-generating view, if we have high-dimensional confounding, then poor high-dimension overlap [?]

[3] where the upper bound B_f is actually free of p, while the left side may not. So strict overlap is very 'strict' in high-dimensional setting.

OB - overlap easy to satisfy? an example

Look into the classical setting:

$$e(X) = exp(W_e),\ X \sim N_p(\mu, \Sigma),\ W_e = \beta_0 + \sum_{j=1}^{p} \beta_j X_j \sim N(\beta_0 + \beta^T \mu, \beta^T \Sigma \beta)$$

Does strict overlap hold? for a constant $\eta \in (0, 1)$.
What would influence the divergence $D_f(P_1 || P_0)$?

OB - measure of overlapness*

Overlapness can be a overall concept, or a local concept.

- generally, how strong tails of $e(x)$
- $min(\eta)$ that satisfy $\eta < e(X) < 1 - \eta$ (intuited by strict overlap)
- $E(e(x)(1 - e(x))$(a bounded measure) ?
- (Liu and Li 2025) Let $f_z(u) = Pr(e(X) = u \mid Z = z)$, the Bhattacharyya coefficient

$$\phi \equiv \int_0^1 \sqrt{f_1(u) f_0(u)}\, \mathrm{d}u.$$

- (Kim et al. 2025) Let $\Sigma_z = \mathbb{E}[\phi(x) \otimes \phi(x)\mathbf{1}(Z = z)]$, define weak overlap[?]

$$\Sigma_0 \preceq R\left(\Sigma_1 + \frac{\xi}{n}\mathbf{I}\right) \quad \text{and} \quad \Sigma_1 \preceq R\left(\Sigma_0 + \frac{\xi}{n}\mathbf{I}\right)$$

- (Ma, Sasaki, and Wang 2025) a test of overlap(tail index) using most extreme ps scores

OB - measure of balance

Balance is an overall concept(no local version), and we need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[4]); mean-matching/pair-matching (Rubin 1973)
- Mahalanobis distance[5] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[4] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[5] MD does not perform well when covariates have nonellipsoidal distributions?

review of overlap/balance-related methods

overlap/balance-related methods*

- trimming(Crump et al. 2009) or ps score truncation(ad hoc)
- (Li, L. E. Thomas, and Li 2019) [ATE over target population] overlap weights, with tilting $h(x) = e(x)(1 - e(x))$, finite exact balance achieved by maximum likelihood under logistic regression

$$w_1(x) \propto 1 - e(x), w_0(x) \propto e(x)$$

- (Imai and Ratkovic 2014) (Athey, Imbens, and Wager 2018) Covariate balancing propensity score, optimize logistic model subject to covariate balancing constraints[6]
- (Diamond and Sekhon 2013) GenMatch searches distance metrics to find the measure that optimizes post-matching covariate balance.
- (Hainmueller 2012) Entropy balancing reweighting

[6] the unbiasedness of IPW is due to balancing property $E(XZ/e(X)) = E(X(1 - Z)/(1 - e(X))$

overlap/balance-related methods - more on CRF*

(Wager and Athey 2018)

- Bias of estimation related to overlap

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

- Tree beats NN under settings of low-dimensional signal structure. So if poor overlap in a high-order-interaction-terms sense, hard to grow a tree?
- CRF is a data-driven way to determine which nearby observations receive more weight. Ideal(?) tree is ideal weights.
- More questions
  - is strict overlap plausible for tree? how small can η be?
  - how does sample balance influence the structure(depth, t/c sample ratio in leaves, overall balance of tree(?)[7]), and then the performance of the grown tree?
  - could match/balance based weighting be used in CRF? (e.g. a balance-aware way of sample weighting/splitting[8])

[7] look at the assumptions

[8] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

overlap/balance-related methods

test then pool/weight?

OB+RCT, matching weighting, simulation, etc

OB - matching and weighting

Connection to Matching and Regression

- Matching: link "similar" cases in two samples, discard unmatched cases (bottom-up approach)
- Weighting: apply weights to entire samples, designed to create global balance (top-down approach)
- Intrinsic connection: Overlap weighting approaches many-to-many matching as the propensity score model becomes increasingly complex
- Why? The limit of a PS model is a saturated model with a fixed effect (dummy variable) for each design point (or small neighborhood, for continuous variables)

Connection to Matching and Regression

- Let the sample count for x_i in group $z = 0, 1$ be N_{zi}, the estimated PS from the saturated model is $\hat{e}(x_i) = N_{1i}/(N_{0i} + N_{1i})$
- The overlap weight for trt and con units is $N_{0i}/(N_{0i} + N_{1i})$ and $N_{1i}/(N_{0i} + N_{1i})$, respectively
- In many-to-many matching: at each design point x, each trt unit is matched to the N_{0i} controls at the same x, each control to N_{1i} trt units – exactly the same as the overlap weighting
- The total overlap weight for each design point x and hence of $\hat{\tau}(x_i)$ is $N_{0i}N_{1i}/(N_{0i} + N_{1i})$
- This is exactly the precision weight attached to $\bar{y}_{1i} - \bar{y}_{0i}$ in the fixed-effects for each design point OLS outcome model $Y_{zi} = \alpha_i + z\tau + \epsilon_{zi}$

- (Lin, Ding, and Han 2023)

combine rct and ob(why?) - ATE

- naive method: IPW on ob and DIM on rct, and weight average
- ...

combine rct and ob(why?) - CATE*

- simple-but-possibly-wrong method: just combine the two source dataset, and throw it into model
- naive method: separately estimate and average the results.
  - just a linear combination of two estimators, easy to analyze its properties
  - involve tuning[9], not end-to-end
- ad hoc methods: ob for training NN and rct for tuning; ob for growing tree and rct for estimation
- augmentation: overlap/balance-aware augmentation, and throw it into model
  - one can use a smart way
  - analyze its statistical properties?
- end-to-end algorithm: overlap/balance-aware way of splitting in CRF
  - analyze its statistical properties?
- end-to-end algorithm: overlap/balance-aware(matching-based) weighting inside NN model
  - hard to analyze statistical properties

[9] more smart way of tuning?

Simulations settings*

See `https://oftrongs.github.io/Causal/simulation_settings.pdf` for summary of simulation settings in some papers.

Under what kind of settings would our methods be better?(Also Not trivial: the distribution of covariate, and the linearity of outcome model)

N obs data, M rct data; d covariates, within which c are confounders and $d - c$ are prognsotic covariates(assume no instrumental variables)

- $e(x)$, overlap, well-specified or not
- M/N, relative information size of rct sample
- $\mu_0(x), \mu_1(x)$, the response surfaces and CATE $\tau(x)$
- c/d, how many irrelevant covariates
- high-dimensional or not (d/N or c/N, complexity of models)

Summary(before)

- Goal: better estimate CATE
  - data: large-scale Post-OB data and small-scale RCT data
  - problem: poor overlap and severe unbalance
- Method: matching-based
  - data augmentation
  - ITE-imputing loss
- Data source: simulation and real-world data
- Further
  - when would the methods work better(the choice of caliper for matching, the size of ob&rct, the degree of non-overlap, the use of previous algorithm or not, etc)
  - an end-to-end rather than step-wise algorithm
  - multi-treatment

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